1. User

Primary Key: user_id
Foreign Key: Nil

No.	Field name	Datatype (Size)	Key Constraints	Description of the field
1	user_id	Int(11)	Primary Key, Auto_increment	Unique identifier for each user.
2	name	Varchar(100)	Not Null	Stores the full name of the user.
3	email	Varchar(100)	Unique, Not Null	Stores the user's email address for login and communication.
4	password	Varchar(255)	Not Null	Stores the encrypted password of the user.
5	phone	Varchar(15)	Not Null	Stores the user's contact number.
6	role	Enum	Not Null	Specifies whether the user is Parent, LLG, or Admin.
7	status	Enum	Not Null	Indicates whether the user account is Active or Inactive.

No.	Field name	Datatype (Size)	Key Constraints	Description of the field
8	created_at	Timestamp	Not Null	Stores the date and time when the user account was created.

2. Child

Primary Key: child_id
Foreign Key: parent_id references User(user_id)

No.	Field name	Datatype (Size)	Key Constraints	Description of the field
1	child_id	Int(11)	Primary Key, Auto_increment	Unique identifier for each child.
2	parent_id	Int(11)	Foreign Key, Not Null	References the parent user who owns the child profile.
3	name	Varchar(100)	Not Null	Stores the child's full name.
4	age	Int(3)	Not Null	Stores the child's age in years.
5	gender	Enum	Not Null	Stores the child's gender (Male, Female, Other).

No.	Field name	Datatype (Size)	Key Constraints	Description of the field
6	is_flagged	Boolean	Default: False	Indicates whether the child has been flagged for professional review.
7	flag_reason	Varchar(255)	Nullable	Stores the reason why the child was flagged.
8	flagged_at	Timestamp	Nullable	Stores the date and time when the child was flagged.
9	skill_profile	JSON / Object	Not Null	Stores the 6 domain skill scores and progress history.
10	rewards	JSON / Object	Not Null	Stores stars, badges, milestones, and streak data.
11	created_at	Timestamp	Not Null	Stores the date and time when the child profile was created.

3. SkillProfile

Primary Key: skill_profile_id
Foreign Key: child_id references Child(child_id)

No.	Field name	Datatype (Size)	Key Constraints	Description of the field
1	skill_profile_id	Int(11)	Primary Key, Auto_increment	Unique identifier for each skill profile record.
2	child_id	Int(11)	Foreign Key, Not Null	References the child whose skill profile this belongs to.
3	cognitive	Decimal(5,2)	Not Null	Stores the cognitive skill score (0-100).
4	language	Decimal(5,2)	Not Null	Stores the language skill score (0-100).
5	motor	Decimal(5,2)	Not Null	Stores the motor skill score (0-100).
6	social	Decimal(5,2)	Not Null	Stores the social skill score (0-100).
7	emotional	Decimal(5,2)	Not Null	Stores the emotional skill score (0-100).
8	creative	Decimal(5,2)	Not Null	Stores the creative skill score (0-100).

No.	Field name	Datatype (Size)	Key Constraints	Description of the field
9	updated_at	Timestamp	Not Null	Stores the date and time when the skill profile was last updated.

4. VABSAssessment

Primary Key: assessment_id
Foreign Key: child_id references Child(child_id)

No.	Field name	Datatype (Size)	Key Constraints	Description of the field
1	assessment_id	Int(11)	Primary Key, Auto_increment	Unique identifier for each VABS/VSMS assessment record.
2	child_id	Int(11)	Foreign Key, Not Null	References the child who completed the assessment.
3	communication	Decimal(5,2)	Not Null	Stores the communication domain score.
4	daily_living	Decimal(5,2)	Not Null	Stores the daily living skills domain score.

No.	Field name	Datatype (Size)	Key Constraints	Description of the field
5	socialization	Decimal(5,2)	Not Null	Stores the socialization domain score.
6	motor_skills	Decimal(5,2)	Not Null	Stores the motor skills domain score.
7	maladaptive	Decimal(5,2)	Not Null	Stores the maladaptive behavior index score.
8	completed_at	Timestamp	Not Null	Stores the date and time when the assessment was completed.

5. Activity

Primary Key: activity_id
Foreign Key: child_id references Child(child_id)

No.	Field name	Datatype (Size)	Key Constraints	Description of the field
1	activity_id	Int(11)	Primary Key, Auto_increment	Unique identifier for each activity.
2	child_id	Int(11)	Foreign Key, Not Null	References the child for whom the activity is generated.

No.	Field name	Datatype (Size)	Key Constraints	Description of the field
3	title	Varchar(150)	Not Null	Stores the title of the activity.
4	instructions	Text	Not Null	Stores the detailed instructions for the parent.
5	difficulty_level	Int(2)	Not Null	Stores the difficulty level of the activity (1-5).
6	type	Enum	Not Null	Specifies whether the activity is Preset or AI_Generated.
7	created_at	Timestamp	Not Null	Stores the date and time when the activity was created.

6. PresetActivity

Primary Key: preset_activity_id
Foreign Key: activity_id references Activity(activity_id)

No.	Field name	Datatype (Size)	Key Constraints	Description of the field
1	preset_activity_id	Int(11)	Primary Key, Auto_increment	Unique identifier for each preset activity record.

No.	Field name	Datatype (Size)	Key Constraints	Description of the field
2	activity_id	Int(11)	Foreign Key, Not Null	References the base activity this preset extends.
3	category	Varchar(50)	Not Null	Stores the category of the preset activity.
4	is_active	Boolean	Default: True	Indicates whether the preset activity is active and available.

7. AIGeneratedActivity

Primary Key: ai_activity_id
Foreign Key: activity_id references Activity(activity_id)

No.	Field name	Datatype (Size)	Key Constraints	Description of the field
1	ai_activity_id	Int(11)	Primary Key, Auto_increment	Unique identifier for each AI-generated activity record.
2	activity_id	Int(11)	Foreign Key, Not Null	References the base activity this AI-generated extends.
3	generated_from	Varchar(50)	Not Null	Stores the feedback ID or source that triggered generation.

No.	Field name	Datatype (Size)	Key Constraints	Description of the field
4	parent_approved	Boolean	Default: False	Indicates whether the parent has approved the activity.

8. Feedback

Primary Key: feedback_id
Foreign Key: child_id references Child(child_id), activity_id references Activity(activity_id), parent_id references User(user_id)

No.	Field name	Datatype (Size)	Key Constraints	Description of the field
1	feedback_id	Int(11)	Primary Key, Auto_increment	Unique identifier for each feedback submission.
2	child_id	Int(11)	Foreign Key, Not Null	References the child for whom feedback is submitted.
3	activity_id	Int(11)	Foreign Key, Not Null	References the activity for which feedback is submitted.
4	parent_id	Int(11)	Foreign Key, Not Null	References the parent who submitted the feedback.

No.	Field name	Datatype (Size)	Key Constraints	Description of the field
5	correct	Boolean	Not Null	Indicates whether the child answered correctly.
6	time_taken	Int(5)	Not Null	Stores the time taken to complete the activity (in seconds).
7	difficulty_rating	Enum	Not Null	Stores the parent's rating of difficulty (Easy, Medium, Hard).
8	observations	Text	Nullable	Stores any additional observations from the parent.
9	created_at	Timestamp	Not Null	Stores the date and time when feedback was submitted.

Primary Key: recommendation_id
Foreign Key: child_id references Child(child_id), llg_id references User(user_id)

No.	Field name	Datatype (Size)	Key Constraints	Description of the field
1	recommendation_id	Int(11)	Primary Key, Auto_increment	Unique identifier for each recommendation.

No.	Field name	Datatype (Size)	Key Constraints	Description of the field
2	child_id	Int(11)	Foreign Key, Not Null	References the child receiving the recommendation.
3	llg_id	Int(11)	Foreign Key, Not Null	References the LLG sending the recommendation.
4	message	Text	Not Null	Stores the friendly, non-clinical advice for the parent.
5	status	Enum	Not Null	Indicates whether the message is Sent, Read, or Resolved.
6	created_at	Timestamp	Not Null	Stores the date and time when the recommendation was sent.

10. Notification

Primary Key: notification_id
Foreign Key: user_id references User(user_id)

No.	Field name	Datatype (Size)	Key Constraints	Description of the field
1	notification_id	Int(11)	Primary Key, Auto_increment	Unique identifier for each notification record.
2	user_id	Int(11)	Foreign Key, Not Null	References the user receiving the notification.
3	type	Enum	Not Null	Specifies the type of notification (Flag_Alert, Activity_Ready, Recommendation, Reminder).
4	message	Varchar(255)	Not Null	Stores the notification message.
5	is_read	Boolean	Default: False	Indicates whether the notification has been read by the user.
6	created_at	Timestamp	Not Null	Stores the date and time when the notification was sent.

NORMALIZATION

First Normal Form (1NF)

A table is in First Normal Form (1NF) if:

- All values in each column are atomic (indivisible)
- Each column contains only a single type of data
- Each row is uniquely identifiable, meaning there is a primary key

Example (Before 1NF — Unnormalized Table):

Child Table (Unnormalized)

Child_ID	Child_Name	Age	Activities	Rewards
C001	Aarav	4	Animal Sounds, Shape Hunt, Color Sort	Stars: 42, Badges: First Activity
C002	Riya	5	Story Time, Number Game	Stars: 28, Badges: Reading Star

Problems:

- Activities column contains multiple values (not atomic)
- Rewards column contains multiple values (stars and badges together)
- Violates 1NF

Conversion to 1NF:

Child Table (After 1NF)

Child_ID	Child_Name	Age	Activity	Stars	Badges
C001	Aarav	4	Animal Sounds	42	First Activity
C001	Aarav	4	Shape Hunt	42	First Activity
C001	Aarav	4	Color Sort	42	First Activity
C002	Riya	5	Story Time	28	Reading Star
C002	Riya	5	Number Game	28	Reading Star

Now: Each cell has a single value. But there is still redundancy (Child_Name, Age, Stars, Badges repeated).

Second Normal Form (2NF)

A table is in Second Normal Form (2NF) if:

- It is already in 1NF
- It removes partial dependencies — all non-key attributes must be fully dependent on the entire primary key
- If the table has a composite primary key, non-key attributes should not be dependent on just a part of the primary key

Example (Before 2NF — 1NF Table):

Child Activity Table (1NF)

Child_ID	Child_Name	Age	Activity	Stars	Badges
C001	Aarav	4	Animal Sounds	42	First Activity
C001	Aarav	4	Shape Hunt	42	First Activity
C001	Aarav	4	Color Sort	42	First Activity
C002	Riya	5	Story Time	28	Reading Star
C002	Riya	5	Number Game	28	Reading Star

Composite Primary Key: (Child_ID, Activity)

Problem: Child_Name, Age, Stars, and Badges depend only on Child_ID — not on the full composite key (Child_ID + Activity). This is a partial dependency.

Conversion to 2NF:

Child Table

Child_ID	Child_Name	Age	Stars	Badges
C001	Aarav	4	42	First Activity
C002	Riya	5	28	Reading Star

Activity Table

Child_ID	Activity
C001	Animal Sounds
C001	Shape Hunt
C001	Color Sort
C002	Story Time
C002	Number Game

Now: Child_Name, Age, Stars, and Badges depend on Child_ID (the whole primary key). Activity depends on Child_ID as well, but is now in a separate table.

Third Normal Form (3NF)

A table is in Third Normal Form (3NF) if:

- It is already in 2NF
- It removes transitive dependencies — non-key attributes should not depend on other non-key attributes
- All attributes must be directly dependent on the primary key

Example (Before 3NF — 2NF Tables):

Child Table (2NF)

Child_ID	Child_Name	Age	DOB	Stars	Badges	Flag_Reason
C001	Aarav	4	2022-06-15	42	First Activity	Motor skills: 35%
C002	Riya	5	2021-03-10	28	Reading Star	NULL

Enrollment Table (2NF)

Child_ID	Activity
C001	Animal Sounds
C001	Shape Hunt
C002	Story Time

Problem: Age depends on DOB (transitive dependency). If DOB changes, Age becomes outdated.

Conversion to 3NF:

Child Table

Child_ID	Child_Name	DOB	Stars	Badges	Flag_Reason
C001	Aarav	2022-06-15	42	First Activity	Motor skills: 35%
C002	Riya	2021-03-10	28	Reading Star	NULL

Activity Table

Child_ID	Activity
C001	Animal Sounds
C001	Shape Hunt
C002	Story Time

Now: Age is removed because it depends on DOB. Age can be calculated from DOB whenever needed — no transitive dependency.